Experiment A Analysis

This notebook reproduces the main statistical analysis for Experiment A.

Statistical Summary

Primary hypothesis testing methods:

Experiment A:

- Binomial test
- Confidence interval
- Cohen's h

Experiment B:

- Welch t-test
- Chi-square test
- Cramér's V
- Cohen's d

Files used

- `experiment_a_data_participant_summary.csv`
- `experiment_a_data_scenario.csv`

Main analyses

1. Binomial test on final participant preference
2. 95% confidence interval for aligned preference proportion
3. Cohen's h for practical significance
4. Paired t-test on mean aligned vs generic scores
5. Cohen's d for paired mean difference
6. Persona-level summary
7. Scenario-level summary
8. Visualization export

In [5]:
```
import pandas as pd
import numpy as np
import math
from scipy import stats
import matplotlib.pyplot as plt
from pathlib import Path

pd.options.display.float_format = '{:.4f}'.format
```

In [6]:
```
base = Path("")

part = pd.read_csv(base / "../data/experiment_a_data_participant_summary.csv", sep=",", decimal=".", encoding="utf-8-sig")
part.head()
```

Out[6]:

	Participant_ID	Persona	Form_Version	Aligned_Count	Generic_Count	Total_Scenarios	Aligned_Rate	Final_Preference	Mean_Aligned_Score	Mean_Generic_Score	Difference_Score	Overall_Preference	Preferred_Aligned
0	P001	STJ	A	6	0	6	1.0000	Aligned	4.5833	3.0833	1.5000	Structured/personalized responses	1
1	P002	STJ	A	0	6	6	0.0000	Generic	3.1250	2.5417	0.5833	Structured/personalized responses	0
2	P003	STJ	A	6	0	6	1.0000	Aligned	5.0000	3.0000	2.0000	No preference	1
3	P004	STJ	A	5	1	6	0.8333	Aligned	4.4167	2.8750	1.5417	Generic responses	1
4	P005	STJ	A	5	1	6	0.8333	Aligned	4.4167	3.8750	0.5417	Generic responses	1

In [7]:
```
scen = pd.read_csv(base / "../data/experiment_a_data_scenario.csv", sep=",", decimal=".", encoding="utf-8-sig")
scen.head()
```

Out[7]:

	Participant_ID	Scenario_ID	Persona	Form_Version	Aligned_Response	Clarity_A	Clarity_B	Satisfaction_A	Satisfaction_B	Trust_A	...	Clarity_Aligned	Clarity_Generic	Satisfaction_Aligned	Satisfaction_Generic	Trust_Aligned	Trust_Generic	Reuse_Aligned	Reuse_Generic	Mean_Aligned	Mean_Generic
0	P001	S1_Student_Exchange	STJ	A	A	5	4	5	3	4	...	5	4	5	3	4	3	5	4	4.7500	3.5000
1	P001	S2_Tuition_Payment	STJ	A	A	5	4	4	2	4	...	5	4	4	2	4	3	5	3	4.5000	3.0000
2	P001	S3_Wi-Fi_Access	STJ	A	A	5	2	4	2	4	...	5	2	4	2	4	3	5	3	4.5000	2.5000
3	P001	S4_Suspicious_Email	STJ	A	A	5	2	5	3	5	...	5	2	5	3	5	3	5	4	5.0000	3.0000
4	P001	S5_Attendance_Policy	STJ	A	A	5	3	4	3	4	...	5	3	4	3	4	4	4	3	4.2500	3.2500

5 rows × 25 columns

In [8]:
```
# Core Experiment A calculations
n = len(part)
aligned_n = int((part["Final_Preference"] == "Aligned").sum())
generic_n = int((part["Final_Preference"] == "Generic").sum())
p_hat = aligned_n / n

# Binomial test
binom_res = stats.binomtest(aligned_n, n=n, p=0.5, alternative="greater")
ci = binom_res.proportion_ci(confidence_level=0.95, method="wilson")
cohens_h = 2 * (math.asin(math.sqrt(p_hat)) - math.asin(math.sqrt(0.5)))

mean_aligned = part["Mean_Aligned_Score"].mean()
mean_generic = part["Mean_Generic_Score"].mean()
mean_diff = part["Difference_Score"].mean()

sd_diff = part["Difference_Score"].std(ddof=1)
se_diff = sd_diff / math.sqrt(n)

t_res = stats.ttest_rel(part["Mean_Aligned_Score"], part["Mean_Generic_Score"])
t_crit = stats.t.ppf(0.975, df=n-1)

wilcoxon_res = stats.wilcoxon(part["Mean_Aligned_Score"], part["Mean_Generic_Score"], alternative="greater")

ci_diff_low = mean_diff - t_crit * se_diff
ci_diff_high = mean_diff + t_crit * se_diff

cohens_d_paired = mean_diff / sd_diff

summary = pd.DataFrame({
    "Metric": [
        "Participants",
        "Aligned preference count",
        "Generic preference count",
        "Aligned preference proportion",
        "Binomial test p-value",
        "95% CI lower",
        "95% CI upper",
        "Mean aligned score",
        "Mean generic score",
        "Mean difference",
        "Paired t statistic",
        "Paired t-test p-value",
        "Wilcoxon statistic",
        "Wilcoxon p-value",
        "95% CI diff lower",
        "95% CI diff upper",
        "Cohen's h",
        "Cohen's d (paired)"
    ],
    "Value": [
        n,
        aligned_n,
        generic_n,
        p_hat,
        binom_res.pvalue,
        ci.low,
        ci.high,
        mean_aligned,
        mean_generic,
        mean_diff,
        t_res.statistic,
        t_res.pvalue,
        wilcoxon_res.statistic,
        wilcoxon_res.pvalue,
        ci_diff_low,
        ci_diff_high,
        cohens_h,
        cohens_d_paired
    ]
})

summary
```

Out[8]:

	Metric	Value
0	Participants	870000
1	Aligned preference count	42.0000
2	Generic preference count	15.0000
3	Aligned preference proportion	0.7368
4	Binomial test p-value	0.0002
5	95% CI lower	0.6318
6	95% CI upper	1.0000
7	Mean aligned score	4.2939
8	Mean generic score	3.3655
9	Mean difference	0.9284
10	Paired t statistic	7.5778
11	Paired t-test p-value	0.0000
12	Wilcoxon statistic	1665.5000
13	Wilcoxon p-value	0.0000
14	95% CI diff lower	0.6829
15	95% CI diff upper	1.1738
16	Cohen's h	0.4935
17	Cohen's d (paired)	1.0037

In [9]:
```
# Persona-level summary
persona_summary = (
    part.groupby("Persona", dropna=False)
    .agg(
        n=("Participant_ID", "count"),
        aligned_count=("Final_Preference", lambda s: (s == "Aligned").sum()),
        mean_aligned=("Mean_Aligned_Score", "mean"),
        mean_generic=("Mean_Generic_Score", "mean"),
        mean_difference=("Difference_Score", "mean"),
    )
    .reset_index()
)
persona_summary["aligned_rate"] = persona_summary["aligned_count"] / persona_summary["n"]
persona_summary
```

Out[9]:

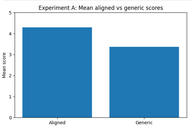
	Persona	n	aligned_count	mean_aligned	mean_generic	mean_difference	aligned_rate
0	NFJ	12	8	3.7153	3.1458	0.5694	0.6667
1	NTP	12	9	4.5868	3.6632	0.9236	0.7500
2	SFP	11	8	4.1477	3.2424	0.9053	0.7273
3	STJ	22	17	4.6227	3.3845	1.1383	0.7727

In [10]:
```
# Scenario-level summary
scenario_summary = (
    scen.groupby("Scenario_ID", dropna=False)
    .agg(
        n=("Participant_ID", "count"),
        mean_aligned=("Mean_Aligned", "mean"),
        mean_generic=("Mean_Generic", "mean"),
    )
    .reset_index()
)
scenario_summary["mean_difference"] = scenario_summary["mean_aligned"] - scenario_summary["mean_generic"]
scenario_summary
```

Out[10]:

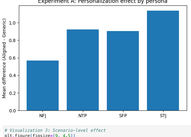
	Scenario_ID	n	mean_aligned	mean_generic	mean_difference
0	S1_Student_Exchange	57	4.8211	2.9649	1.4561
1	S2_Tuition_Payment	57	4.3916	3.2632	1.0482
2	S3_Wi-Fi_Access	57	4.3860	3.2456	1.1404
3	S4_Suspicious_Email	57	4.3070	3.6535	0.6535
4	S5_Attendance_Policy	57	4.2281	3.3333	0.8947
5	S6_Elective_Courses	57	4.1053	3.6711	0.4342

In [11]:
```
# Visualization 1: Mean aligned vs generic scores
plt.figure(figsize=(6, 4))
plt.bar(["Aligned", "Generic"], [mean_aligned, mean_generic])
plt.ylabel("Mean score")
plt.title("Experiment A: Mean aligned vs generic scores")
plt.ylim(0, 5)
plt.tight_layout()
plt.show()
```



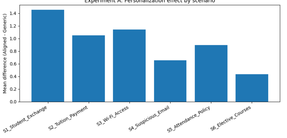
In [12]:
```
# Visualization 2: Persona-level effect
plt.figure(figsize=(6, 4))
plt.bar(persona_summary["Persona"], persona_summary["mean_difference"])
plt.ylabel("Mean difference (Aligned - Generic)")
plt.title("Experiment A: Personalization effect by persona")
plt.tight_layout()

plt.show()
```



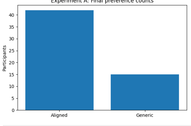
In [13]:
```
# Visualization 3: Scenario-level effect
plt.figure(figsize=(9, 4.5))
plt.bar(scenario_summary["Scenario_ID"], scenario_summary["mean_difference"])
plt.ylabel("Mean difference (Aligned - Generic)")
plt.title("Experiment A: Personalization effect by scenario")
plt.xticks(rotation=30, ha="right")
plt.tight_layout()

plt.show()
```



In [14]:
```
# Visualization 4: Final preference counts
plt.figure(figsize=(6, 4))
plt.bar(["Aligned", "Generic"], [aligned_n, generic_n])
plt.ylabel("Participants")
plt.title("Experiment A: Final preference counts")
plt.tight_layout()

plt.show()
```



In [15]:
```
# Visualization 5: Persona-level comparison
x = np.arange(len(persona_summary["Persona"]))
width = 0.35

plt.figure(figsize=(9, 4.5))
plt.bar(x - width/2, persona_summary["mean_aligned"], width)
plt.bar(x + width/2, persona_summary["mean_generic"], width)
plt.xticks(x, persona_summary["Persona"])
plt.ylabel("Mean Score")
plt.title("Experiment A — Persona-Level Comparison")
plt.legend(["Aligned", "Generic"])
plt.tight_layout()

plt.show()
```

In [16]:
```
# Visualization 6: Scenario-level comparison
x2 = np.arange(len(scenario_summary.index))

plt.figure(figsize=(9, 4.5))
plt.bar(x2 - width/2, scenario_summary["mean_aligned"], width)
plt.bar(x2 + width/2, scenario_summary["mean_generic"], width)
plt.xticks(x2, scenario_summary["Scenario_ID"], rotation=30, ha="right")
plt.ylabel("Mean Score")
plt.title("Experiment A — Scenario-Level Comparison")
plt.legend(["Aligned", "Generic"])
plt.tight_layout()

plt.show()
```